Operations Guide

Surette Nexus · Surette Real Estate, Waterville, Maine

Generated October 27, 2026 from docs/operations-guide.md

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Step-by-step instructions for running, sharing, hosting and maintaining the Surette Nexus website: on a Mac, in Docker, from your Mac with a public link, and on the Oracle Cloud free server. Everything here uses free software. No passwords or keys are written in this guide; where they live is listed in section 2.

1. What runs where

Part	What it does	Port (local)
Website + API (Next.js)	Public site, customer accounts, staff portal (`/admin`), REST API	3100
Background worker	Jobs: market data refresh, AI indexing, emails, clean-up	9464 (health)
Live updates (WebSocket)	Instant refresh in the portal; falls back to Server-Sent Events	3200
PostgreSQL + pgvector	All data, permissions (row-level security), vector search	5434
AI service (Python, `ml/`)	Document Q&A (RAG), semantic search	8600
Ollama	Local AI models (`llama3.2:3b` , `mxbai-embed-large`)	11434
Mailpit	Test inbox for sign-up and password-reset emails	8025
Market Lab dashboard (Streamlit)	Analyst view of the models	8501

Where	Address
Live demo (Oracle)	https://129-213-166-31.sslip.io
Local website	http://localhost:3100
Staff portal	`/portal/sign-in` on either address
Customer sign-in	`/sign-in`
Code	https://github.com/AkshaySaraf17/surette-nexus

2. Accounts, keys and private files

Item	Where it is	Notes
Demo logins (local and Oracle)	`DEMO-LOGINS.local.txt` in the project folder	Private, never committed. The passwords in the public code no longer work.
SSH key for the Oracle server	`~/.ssh/surette_oracle` (private) and `~/.ssh/surette_oracle.pub` (public)	Losing the private key means losing access; back it up.
Server secrets (database password, session key)	`~/surette/.env` on the Oracle server	Created on first deploy; never leaves the server.
Local settings	`web/.env` on your Mac	Never committed (git-ignored).
Oracle Cloud account	https://cloud.oracle.com (region US East / Ashburn)	Your login. Never upgrade to Pay As You Go unless you intend to pay.
GitHub	Account `AkshaySaraf17`	Commits use only your name and email.
Oracle API key (optional, for the retry script)	`~/.oci/config` and `~/.oci/oci_api_key.pem`	Only on your Mac.

Never paste passwords, keys or `.env` contents into chats, emails or GitHub.

3. Run it on your Mac (development)

3.1 One-time install

1. Install Homebrew (https://brew.sh), then the free tools:

```
brew install git gh node@22 postgresql@16 python ollama mailpit cloudflared
brew services start ollama
brew services start mailpit
```

2. Get the code:

```
gh auth login
gh repo clone AkshaySaraf17/surette-nexus ~/Documents/surette-nexus
cd ~/Documents/surette-nexus/web
cp .env.example .env
```

3. Edit `web/.env` and set at least `JWT_SECRET` (a long random value from `openssl rand -base64 48`). For the AI features also set `OLLAMA_URL="http://localhost:11434"` and `ML_SERVICE_URL="http://localhost:8600"`.

4. Install packages, create the database and load demo data:

```
npm install
cd .. && scripts/db.sh start
cd web && npm run db:migrate && npm run db:seed
```

5. Optional, for the AI features (about 4 GB of downloads):

```
cd ~/Documents/surette-nexus && scripts/ml-setup.sh
```

3.2 Start and stop

```
cd ~/Documents/surette-nexus && scripts/dev.sh
```

This starts the database, website, worker, live updates and (if set up) the AI service. Open http://localhost:3100. Press **Ctrl+C** to stop. In VS Code: **Terminal** → **Run Build Task** (Cmd+Shift+B) runs "Start everything".

3.3 Demo logins

Seeding creates an owner, an agent and a customer account (`owner@surette.test`, `agent@surette.test`, `client@surette.test`). Their current passwords are in `DEMO-LOGINS.local.txt`. After a fresh `npm run db:seed`, the passwords printed by the seed apply again; change them before sharing the site (section 9.3).

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4. Settings (`web/.env`)

Setting	Purpose	Typical value
`DATABASE_URL`	PostgreSQL connection	`postgresql://surette@localhost:5434/surette`
`JWT_SECRET`	Signs sign-in sessions (keep secret, long, random)	`openssl rand -base64 48`

Setting	Purpose	Typical value
`PUBLIC_SITE_URL`	The site's public address (links in emails, Google sign-in)	`http://localhost:3100` locally
`SITE_TEAM_SLUG`	Which team the site shows	`surette`
`STORAGE_DIR`	Where uploaded photos and documents are stored	`.storage`
`OLLAMA_URL`, `OLLAMA_MODEL`	Local AI for nicer assistant wording	`http://localhost:11434`, `llama3.2:3b`
`ML_SERVICE_URL`, `ML_SERVICE_TOKEN`	AI service for document Q&A and semantic search	`http://localhost:8600`, empty locally
`OLLAMA_EMBED_MODEL`	Embedding model for vector search	`mxbai-embed-large`
`ACCOUNT_EMAILS`	`off` = no emails: sign-ups confirmed at once, "Forgot password?" hidden	empty locally, `off` on the demo server
`SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASS`, `MAIL_FROM`	Real email sending (any provider)	empty = Mailpit locally
`GOOGLE_CLIENT_ID`, `GOOGLE_CLIENT_SECRET`	"Continue with Google" (hidden when empty)	from Google Cloud Console
`METRICS_TOKEN`	Protects `/api/metrics` (required in production)	random
`NEXT_PUBLIC_REALTIME_URL`	WebSocket address for live updates (empty = Server-Sent Events)	`ws://localhost:3200/ws` with `dev.sh`

Setting	Purpose	Typical value
`NEXT_PUBLIC_SITE_DEMO`	Demo label (badge, footer, no search indexing); `false` removes it	default on
`REDIS_URL`	Optional shared cache and rate limits	empty
`CORS_ORIGINS`	Other sites allowed to call the API	`http://localhost:8081`

`NEXT_PUBLIC_*` values are built into the website, so changing them needs a rebuild.

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5. Docker on your Mac (whole stack in containers)

1. Free container runtime: `brew install colima docker docker-compose docker-buildx`, then `colima start`.
2. Start everything (creates a `.env` with random secrets on first run, loads demo data):

   ```
   scripts/docker-demo.sh
   ```

   Open http://localhost:8080. Stop with `scripts/docker-demo.sh stop`.
3. Extras: `docker compose --profile observability up -d` (Prometheus, Grafana, Loki); `docker compose --profile ai up -d` (Ollama + AI service; set `ML_SERVICE_URL=http://ml:8600` and `ML_SERVICE_TOKEN` in `.env`).
4. **Disk warning:** Colima's virtual disk grows and can fill a small Mac disk. Stop it when not in use with `colima stop`, and check space with `df -h /`.

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6. Share from your Mac with a public link (free, no account)

```
scripts/share.sh
```

It builds the production site, starts it with the worker and AI service, and prints a `https://….trycloudflare.com` link. Every feature works, including local AI. The link works only while

the Mac is awake and the script runs, and it changes every time. Use `scripts/share.sh --rebuild` after code changes.

7. Oracle Cloud free hosting

7.1 What's deployed now

Item	Value
Server	`VM.Standard.E2.1.Micro` (1 CPU, 1 GB memory + 4 GB swap), Ubuntu 24.04, Ashburn AD-1
Public IP	`129.213.166.31`
Address	https://129-213-166-31.sslip.io (free name from sslip.io; HTTPS by Caddy + Let's Encrypt)
Network	VCN `surette-vcn`, public subnet; ingress rules TCP 22, 80, 443
On the server	`~/surette/docker-compose.yml`, `Caddyfile`, `.env` (secrets)
Containers	`db` (PostgreSQL 17 + pgvector), `web`, `worker`, `caddy`
Site image	`ghcr.io/akshaysaraf17/surette-nexus:main` (built by GitHub on every push to main; public)
Not included	Ollama and the AI service (need ~8 GB): the assistant uses built-in answers, search uses keywords

7.2 Creating a server (for reference or a new one)

1. Sign in at https://cloud.oracle.com and check the region (top right) is **US East (Ashburn)**.
2. Network: https://cloud.oracle.com/networking/vcns → **Actions** → **Start VCN Wizard** → **Create VCN with Internet Connectivity** → name `surette-vcn` → **Next** → **Create**.
3. Server: https://cloud.oracle.com/compute/instances/create
    - Image: **Canonical Ubuntu 24.04**.
    - Shape: **VM.Standard.A1.Flex** (Ampere; 2 OCPU / 12 GB is enough for the full AI) or **VM.Standard.E2.1.Micro** (always available in AD-1; no AI).
    - Networking: existing VCN `surette-vcn`, subnet **public subnet-surette-vcn**, **assign a public IPv4 address: on**.
    - SSH keys: **Paste public keys** → the contents of `~/.ssh/surette_oracle.pub`.
    - Boot volume: default for the Micro; 100 GB for an A1. The free allowance is 200 GB in total.
    - **Create.** "Out of capacity" for A1 is common: try AD-1, AD-2, AD-3, a smaller size, or another time.
4. Firewall: instance → **Networking** → subnet → **Security** → **Default Security List for surette-vcn** → **Add Ingress Rules**: Source `0.0.0.0/0`, TCP, destination ports `80` and `443`.
5. Prepare the server (swap, Ubuntu firewall, Docker). Replace the IP:

```
ssh -i ~/.ssh/surette_oracle ubuntu@129.213.166.31
sudo fallocate -l 4G /swapfile && sudo chmod 600 /swapfile && sudo mkswap /swapfile
echo "/swapfile none swap sw 0 0" | sudo tee -a /etc/fstab
sudo iptables -I INPUT 5 -p tcp --dport 80 -m state --state NEW -j ACCEPT
sudo iptables -I INPUT 5 -p tcp --dport 443 -m state --state NEW -j ACCEPT
sudo sh -c "iptables-save > /etc/iptables/rules.v4"
sudo apt-get update && sudo apt-get install -y docker.io docker-compose-v2
sudo usermod -aG docker ubuntu && exit
```

7.3 Deploy and update

From your Mac, after GitHub's checks pass on main:

```
scripts/deploy-oracle.sh 129.213.166.31
```

It copies `deploy/oracle/*` to the server, creates secrets there on the first run, downloads the latest site image, applies database migrations, loads demo data and sets new random demo passwords on the first run only (saved to `DEMO-LOGINS.local.txt`), starts the containers and waits for HTTPS.

7.4 Everyday server commands

Connect: `ssh -i ~/.ssh/surette_oracle ubuntu@129.213.166.31`, then `cd ~/surette`.

Task	Command (on the server, in `~/surette`)
Status	`docker compose ps`
Logs (last 100 lines)	`docker compose logs --tail 100 web worker`
Restart the site	`docker compose restart web worker`
Stop / start everything	`docker compose down` / `docker compose up -d`
Memory use	`free -m` and `docker stats --no-stream`
Disk use	`df -h /`

Backup (run on your Mac; creates a compressed copy of the database):

```
ssh -i ~/.ssh/surette_oracle ubuntu@129.213.166.31 'cd surette && docker compose exec -T
```

Restore a backup into the server (replaces the data):

```
gunzip -c surette-backup-YYYY-MM-DD.sql.gz | ssh -i ~/.ssh/surette_oracle ubuntu@129.213.
```

Uploaded photos and documents live in the `storage` Docker volume on the server.

7.5 A custom name: surette-demo.duckdns.org (free)

1. Go to https://www.duckdns.org and **sign in with GitHub**.
2. Type `surette-demo` (or another free name), click **add domain**, enter the server IP in **current ip**, click **update ip**.
3. Switch the site to the new name (Caddy gets a new HTTPS certificate automatically):

```
ssh -i ~/.ssh/surette_oracle ubuntu@129.213.166.31 "cd surette && sed -i 's/^SITE_H(
```

4. Open https://surette-demo.duckdns.org (allow a minute for the certificate). If you set up Google sign-in, add the new address to its authorized redirect URIs.

A `.com` name costs about $10–12 a year (e.g. Cloudflare Registrar or Porkbun); point its DNS A record to the server IP and use the same `SITE_HOST` change.

7.6 Moving to a bigger free server (full AI online)

When an A1 server is available (2 OCPU / 12 GB or more), create it as in 7.2, prepare it, take a backup of the current database, deploy with `scripts/deploy-oracle.sh <new-ip>`, restore the backup, add the AI services (`docker compose --profile ai`), then point the name (sslip or DuckDNS) to the new IP.

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8. GitHub and automatic checks

- Every push to `main` runs: lint and type checks, unit and database tests, browser tests (with a test email inbox), the Python AI-service tests, a container build with a security scan, and publishes the site image to GitHub Packages.
- See results: https://github.com/AkshaySaraf17/surette-nexus/actions.
- Update the live site only after the checks are green (section 7.3).
- Commits are made with your name only. Run `gitleaks git --staged` before pushing to catch secrets.

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9. Common tasks

9.1 Run the tests

```
cd web && npm test                 # unit + database tests
cd web && npm run test:e2e         # browser tests (with scripts/dev.sh running)
cd ml && .venv/bin/python -m pytest
```

9.2 Rebuild the Market Lab results

```
cd ml && .venv/bin/python -m market_lab.build
cd ml && PYTHONPATH=. .venv/bin/streamlit run market_lab/dashboard.py
```

9.3 Change the demo passwords

- **New staff accounts:** the owner adds them in **Portal** → **Team & accounts** with a starting password.
- **Resetting a password:** the owner account can reset any account's password through the admin API (`PATCH /api/admin/users/<id>` with `{"newPassword": "…"}`, at least 12 characters with letters and numbers). There's no button for it in the portal yet.
- **Customers** reset their own password with "Forgot password?" when emails are switched on.
- **Server:** a fresh database (empty `pgdata` volume) gets new random demo passwords on the next deploy.

9.4 Remove the demo label

Build with `NEXT_PUBLIC_SITE_DEMO=false` (Docker: `SITE_DEMO=false` in `.env`, then rebuild).

10. Troubleshooting

Problem	Fix
Pages flash / reload constantly and won't scroll (local)	Stop the site, delete the build cache `rm -rf web/.next`, start again.
Mac disk full	`colima stop`; `npm cache clean --force`; check `df -h /`.
"Port already in use"	Something is still running: stop `scripts/dev.sh` (Ctrl+C) or find it with `lsof -i :3100`.
Sign-in says "too many attempts"	Rate limit: wait 15 minutes.
Oracle "Out of capacity"	Try other availability domains, a smaller size, another time, or the Micro shape.
Live site shows 502 right after deploy	The site is still starting: wait 30–60 seconds. Check `docker compose logs --tail 50 web`.
HTTPS certificate not issued	Ports 80/443 must be open in both the Oracle security list and the server's iptables (7.2).
Live site slow on first visit	Normal on the 1 GB server after a quiet period; next pages load in about a second.
AI assistant gives plain answers	Expected on the small server (no local AI), or when the AI's wording failed its safety check.

Problem	Fix
Cloudflare link stopped working	The Mac slept or `share.sh` stopped; run it again for a new link.

11. Security checklist

- Never commit `web/.env`, `DEMO-LOGINS.local.txt`, `~/.oci/*` or the SSH private key.
- Change demo passwords before sharing a site; send staff logins privately.
- Keep the Oracle account on the Always Free tier (no Pay As You Go) to avoid charges.
- The site is labelled as a demo and asks search engines not to index it.